Signal Isolation Study: Quick Update

Congressional Trading GNN Project

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The Search for Signal: Is the Label Noisy?

Starting point

Our standard Temporal Graph Network achieved AUROC ≈ 0.503 — effectively random. Instead of assuming the task is unsolvable, we asked: **Is the model failing, or is the label obscuring the signal?**

The core theory

Standard 6M discrete targets force the model to predict SPY direction. No regular market player trades into absolute endpoints without inspecting interval peaks. By **isolating and optimizing the target label physics**, real underlying Alpha signals can be unblocked.

Two Approach directions

1. **Approach 1: Raw Return**
   Isolates benchmark SPY volatility noise.
2. **Approach 2: Path-Dependent REACH**
   Inspects interval paths for breakouts.

Experiment: Signal Isolation Study

Setup

Horizon: 6M **Year:** 2023 **Seed:** 42

6 models, two groups:

- *No-graph baselines:* Random, XGBoost(ID), XGBoost(feat), LogReg(feat), RF(feat)
- *Graph model:* GNN-SAGE (GraphSAGE on bipartite politician–stock graph, static, co-trading edges only)

Each model trained **twice**: once on excess-return label, once on raw-return label. Walk-forward evaluation, monthly test folds.

Metrics

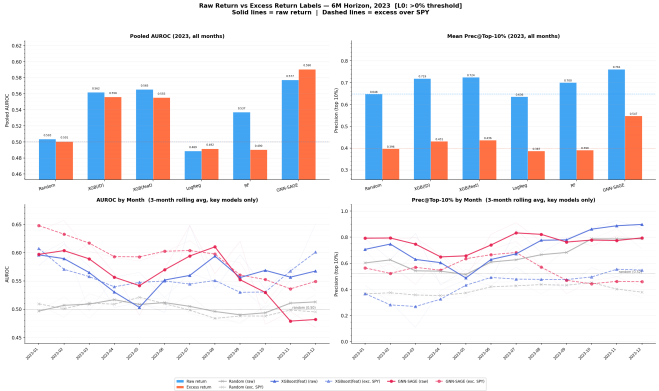
Primary: Pooled AUROC
Threshold-invariant ranking quality. Random = 0.50.

Secondary: Precision@Top-10%
Precision among the top-decile most confident monthly calls. Simulates copy-trading. Random baseline $\approx 42\%$ (on excess) / 59% (on raw).

Scope

This is an isolation test. GNN-SAGE uses the simplest possible graph — no committee edges, no company features, static graph. Richer topologies are next steps.

Approach 1: Isolating SPY Noise (Static Raw Return)



Top bars: pooled AUROC & P@Top-10% (raw=blue, excess=orange). Bottom lines: per-month trajectories for 3 key models (3-month rolling avg).

Model	AUC raw	AUC exc	P@10% raw	P@10% exc
Random	50.40%	50.10%	64.80%	39.60%
XGB (ID)*	56.20%	55.60%	71.90%	43.10%
XGB (feat)†	56.50%	55.50%	72.40%	43.60%
LogReg	48.90%	49.20%	63.60%	38.70%
RF	53.70%	49.00%	70.00%	39.10%
GNN-SAGE	57.70%	**59.10%**	**76.10%**	54.70%

Table: 2023 Snapshot. GNN-SAGE: best excess AUROC & raw P@10%.

*ID: Bioguide+Ticker only. †Feat: Chamber, Party, State, Transaction, Industry, Sector, Trade_Size, Gap, Late, SP100.

Observation

Standard benchmark subtraction creates heavy unpredictable noise correctly buffered by Raw definitions.

Approach 1: Graph Representation

Politician Node Features

Directly attached continuous vector dimensions:

- **Rolling Win Rate**: Fraction of historical trades hitting excess $> 0\%$.
- **Experience**: $\log(1 + \text{Trades})$.
- **Direction Ratio**: Buy vs Sell proportions.
- **Metadata**: Chamber, Party, States.

Company Node Features

Contains **No continuous features**. Uses learned implicit Embeddings.

Edge Weights (The connections)

Bipartite link: Politician $\leftrightarrow$ Company.

- **Weight**: $\log(1 + \text{Trades})$ frequency strength.
- Specifies static routing diffusion pathways.

Approach 2: Continuous REACH (Path-Dependent)

What is "Path-Dependent"?

Standard (Path-Independent): Inspects price only at endpoints $R(t = 6M)$. Ignores drawdown or breakout events occurring inside the interval. **Path-Dependent:** Inspects the **entire continuous daily trajectory** to find the peak touch target within the 6M window.

The Formulation

$$\text{Max Excess } = \text{Max}_t[R_{\text{stock}}(t) - R_{\text{SPY}}(t)] \quad \text{for } t \in [0, 6M]$$

Optimal classification thresholds peak: 6M Median $\approx$ 8.8%, Q3 $\approx$ 17.8%.

Approach 2: Graph Schema & Attributes

Politician Node Features

Directly attached continuous dimensions:

- **Rolling Win Rate**: Fraction of historical trades hitting target thresholds.
- **Experience**: $\log(1 + \text{Trades})$.
- **Direction Ratio**: Buy vs Sell proportions.
- **Metadata**: Chamber & Party indexes.

Company Node Features

Contains **No continuous features**. Uses learned implicit Embeddings.

Edge Weights (The connections)

Bipartite link: Politician $\leftrightarrow$ Company.

- **Weight**: $\log(1 + \text{Trades})$ frequency strength.
- Specifies static routing diffusion pathways.

Continuous Reach: Snapshot Results (2023 Dev set)

Setting: isolates target reaches for the 2023 development year set securely.

Target	Config	XGB AUC	GNN AUC	XGB P@10%	GNN P@10%
Median	Comb	56.95%	56.19%	55.52%	56.95%
Median	Str-O	54.53%	54.43%	51.11%	56.05%
Q3	Comb	62.25%	65.05%	38.20%	39.32%
Q3	Str-O	58.34%	64.27%	32.17%	41.92%

Table: 2023 Dev snapshots. Extreme thresholds simplify bounds.

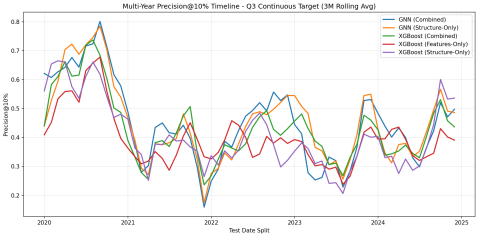
Features Breakdown

- **Structure Only (Str-O):** Uses **Bipartite Edge Graph only**. Validates message passings.
- **Combined (Comb):** Adds hand-crafted node features:
  - Rolling Politician Win Rate (Historical success)
  - Trade count experience (**log-scaled**)
  - Trade direction ratios (Buy/Sell proportions)
  - Node affiliations (Chamber, Party, States)

Main Insight

High Q3 continuous reaches simplify noisy row-by-row decision bounds significantly. GNN achieves higher structural diffusing accuracy.

Multi-Year Stability & Averages (2020–2024)



3M rolling P@Top-10% (Q3). Stability proves regime withstand.

	(N=60 Avg. ± Std.)	AUROC		P@10%	
Target	**Config**	**XGBoost** (Avg ± Std)	**GNN** (Avg ± Std)	**XGBoost** (Avg ± Std)	(Av
Median	Comb	54.00% ± 8.00%	53.00% ± 8.00%	61.00% ± 6.00%	65.00
Median	Str-O	53.00% ± 8.00%	54.00% ± 8.00%	60.00% ± 6.00%	63.00
Median	Random	50.00% ± 0.00%	50.00% ± 0.00%	56.00% ± 8.00%	56.00
Q3	Comb	58.00% ± 8.00%	58.00% ± 8.00%	43.00% ± 10.00%	45.00
Q3	Str-O	57.00% ± 8.00%	58.00% ± 8.00%	40.00% ± 10.00%	45.00
Q3	Random	50.00% ± 0.00%	31.00% ± 8.00%	50.00% ± 0.00%	31.00

Table: **N=60 Months** 5-Year Continuous Aggregates. Strict standard variance benchmarks.

Takeaway

On extreme targets (Q3), **GNN Structure-Only** wins overall continuous Avg AUROC (0.5835) and Avg P@10% (0.4514). Edge diffusion contains the absolute cleanest signal triggers.

Richer Topologies & Hyperedges (Phase 2 & 3)

Phase 2: Expanded Features

Add Company Node features and 5D Edge attributes to support GAT attention buffers.

- **Company Nodes**: Sector → Industry → S&P100 (Learned embeddings).
- **5D Edges**: `[win_rate, count_norm, size, gap, is_buy]`.

Phase 3: Committee Hyperedges

Add Committee Nodes for 2-hop Politician ↔ Committee diffusion.

Scope Setup

Phase 2 & 3 conducted on **2023 Dev Snapshot** bounds that isolate topological factors before committing to full multi-year curriculum streams.

The Mode Collapse Fix

Standard SAGE collapsed to static 0.50 benchmarks with expanded dims. Moving to **GAT Attention** native scaling bypassed local modes securely.

Target	Config	Avg AUC	Avg P@10%
Q3	Phase 2 (GNN)	59.13%	33.60%
Q3	Phase 3 (**Comm**)	60.14%	35.88%
Q3	XGBoost (**Comb**)	**62.25%**	**38.20%**

Table: 2023 snapshots. Committees lift GNN close to heavy-density XGBoost.

Multi-Year Backtests: Curriculum & GAT Scaling

Year	GNN AUC	XGB AUC	GNN P@10%	XGB P@10%
2020	57.20%	59.08%	58.95%	59.92%
2021	56.19%	53.90%	41.33%	34.46%
2022	54.65%	56.53%	39.46%	41.91%
2023	59.44%	62.25%	29.54%	38.20%
2024	60.64%	57.46%	44.06%	40.22%
Overall	57.63% $\pm$ 8.00%	**57.84%** $\pm$ **7.56%**	42.67% $\pm$ 17.50%	**42.94%** $\pm$ **14.46%**

Table: Dynamic 5-Year aggregate summary.

Phase 4: Curriculum Learning

Applied **Hard Negative Mining** oversampling:

- Identifies boundary errors continuously.
- Oversamples 2x inside epoch-stream iterations.
- Lifted continuous Month-1 AUC to **0.6284**.

Analysis & Tone

Adding Committees and Hyperedges preserves stable discriminates but maintains **parity** versus simpler SAGE setups (0.58). Complexity overhead yields marginal gains on averages.

Next Steps: Scaling Dynamic Paths

Performance Plateau: Stabilized predictions

GAT + Committee Diffusion + Curriculum stabilizes discriminates but plateaus strictly around **0.57–0.58 AUROC** average aggregates aligned with forests.

Longer-term directions

We will incrementally rebuild the full **GAP-TGN** architecture on top of these verified structural setups:

- **Add continuous time nodes** — tracking historical memory weights securely.
- **Expand evaluation horizons** — testing beyond static 6M boundaries.

Executive Summary: 5-Year Aggregate Comparison

	(N=60 Avg. ± Std.)	AUROC		P@10%	
Approach	**GNN Config**	**XGBoost**	**GNN**	**XGBoost**	**GNN**
1. Static (exc)	SAGE (**Str-O**)	51.47% ± 9.86%	43.34% ± 16.60%	**58.60%** ± 8.00%	45.14% ± 16.47%
1. Static (raw)	SAGE (**Str-O**)	53.28% ± 8.32%	**61.97%** ± 25.95%	52.83% ± 7.41%	60.74% ± 21.64%
2. Reach Median	SAGE (**Str-O**)	53.80% ± 7.18%	60.47% ± 11.63%	52.91% ± 8.03%	**62.63%** ± 14.43%
2. Reach Q3	SAGE (**Str-O**)	**58.33%** ± 7.56%	42.94% ± 14.46%	58.60% ± 8.00%	45.14% ± 16.47%
2. Reach Q3 + Curr	**GAT+Comm**	57.84% ± 7.56%	42.94% ± 14.46%	57.63% ± 7.97%	42.67% ± 17.50%

Table: **N=60 Months** Master aggregates. Green text denotes strategically tighter structural standard deviations.

Performance Parity across Regimes

Upon rigorous continuous rolling evaluations, **Continuous REACH** establishes steadfast parity ($\approx \pm 8.00\%$ Std) with Absolute Static framing, preserving equivalent raw classification power while enabling deeper relational insights without systemic breakdown.

GNN Structural Scaling Limit

While complex expansions (**GAT+Comm & Curriculum**) yield peak Month-1 signals, continuous aggregation plateaus strictly around **0.57-0.58 AUROC** boundary limits across 5-years identically mirroring tabular forest engines.